

Anomaly Subgraph Detection on Multiple Associated Attributed Networks

Nannan Wu^{ID}, Ying Sun^{ID}, Yazheng Zhao^{ID}, Wenjun Wang^{ID}, Xueli Liu, and Jianxin Li^{ID}

Abstract—With the rapid development of artificial intelligence (AI) technology and large-scale dataset applications, anomaly subgraph detection is becoming an essential and widespread application. Most existing methods detect anomaly subgraphs depending on the observable anomalous attributes in target data. However, they may face challenges when the target data lacks such anomalous attributes. *E.g.*, identifying suspicious gangs in the target transportation graph solely based on road traffic information, or detecting burst events in nonpublic data without explicit anomalous attributes. To address the detection of implicit anomaly subgraphs (IASs) in such target graphs, this article proposes a novel approach IAS detection (IASD) with multidimensional feature transfer. First, our approach is built upon transfer learning techniques and involves fusing features from multiple graphs. Second, we explore a feature extraction procedure based on a graph attention (GAT) network model to generate representations of anomaly features for large multidimensional attribute graphs. Third, we construct a two-layer graph by introducing an attributed graph referred to as a “source graph,” where anomalies can be easily detected, and nodes of the source graph are partially aligned with the target graph. We validate the effectiveness and robustness of IASD through experiments on five real-world datasets, and implement four anomaly subgraph detection tasks in practical applications.

Index Terms—Anomaly alignment, anomaly subgraph detection, complex network, feature transfer learning.

NOMENCLATURE

$N = (G_1, G_2, M)$	Two-layer graph.
$G_1 = (V_1, E_1, X_{G_1})$	Source graph layer.
$G_2 = (V_2, E_2, X_{G_2})$	Target graph layer.
$M \subseteq V_1 \times V_2$	Alignment edge sets.
V_1, V_2 and E_1, E_2	Node sets and edge sets.

Received 20 May 2024; revised 9 October 2024, 1 May 2025, and 24 September 2025; accepted 15 December 2025. This work was supported in part by the Natural Science Foundation of Tianjin under Grant 25JCZDJC01300, in part by the National Natural Science Foundation of China under Grant U2468204, and in part by the Innovative Research Group Project of the National Natural Science Foundation of China under Grant 62272311. (Corresponding author: Ying Sun.)

Nannan Wu is with the School of Cybersecurity, Tianjin University, Tianjin 300350, China (e-mail: nannan.wu@tju.edu.cn).

Ying Sun is with the School of International Business, Zhejiang Yuexiu University, Shaoxing 312000, China (e-mail: yingsun@tju.edu.cn).

Yazheng Zhao and Xueli Liu are with the School of Computer Science and Technology, Tianjin University, Tianjin 300350, China (e-mail: yazhengzhao@tju.edu.cn; xueli@tju.edu.cn).

Wenjun Wang is with the College of Intelligence and Computing, Tianjin University, Tianjin 300350, China (e-mail: wjwang@tju.edu.cn).

Jianxin Li is with the State Key Laboratory of Software Development Environment and Beijing Advanced Innovation Center for Big Data and Brain Computing, Beihang University, Beijing 100191, China (e-mail: lijx@act.buaa.edu.cn).

Digital Object Identifier 10.1109/TNNLS.2025.3647831

$X_{G_1} \in \mathbb{R}^{n \times d_1}$
 $X_{G_2} \in \mathbb{R}^{m \times d_2}$
 $n = |V_1|, m = |V_2|$
 $d_1 \in \mathbb{N}, d_2 \in \mathbb{N}$
 $S_1 = (V_{S_1}, E_{S_1}, X_{S_1})$
 $S_2 = (V_{S_2}, E_{S_2}, X_{S_2})$
 $H \subseteq V_1 \times V_2$
 $x_{G_1}(v) \in \mathbb{R}^{d_1}, v \in V_1$
 $x_{G_2}(v) \in \mathbb{R}^{d_2}, v \in V_2$
 $\mathbf{x}_1 \in \mathbb{R}^n, \mathbf{x}_2 \in \mathbb{R}^m$

Anomaly attribute matrix of G_1 .
 Nonanomaly attribute matrix of G_2 .
 Number of nodes of G_1 and G_2 .
 Dimensionality of node attributes.
 Anomaly subgraph of G_1 .
 Anomaly subgraph of G_2 .
 Prior alignment edge sets.
 Anomaly attribute of G_1 nodes.
 Non-anomaly attribute of G_2 nodes.
 Feature vectors of G_1, G_2 .

I. INTRODUCTION

ANOMALY detection refers to the discovery of entities, events, and observations that deviate from the majority, while anomaly subgraph detection identifies unusual or anomaly subgraphs within a larger graph or network.¹ Anomaly subgraph detection is a vital task in various public safety applications, such as hot spot detection, fraud detection, and intrusion detection.

In general, anomaly subgraph detection specifically focuses on revealing anomalous patterns and facilitating informed decision-making. Most existing approaches achieve this by directly extracting anomaly-related features based on observable anomalous attributes within the target dataset [1], [2]. Nonetheless, they face significant challenges when such anomalous attributes are absent. In practical applications, detecting urban crime hotspots remains a crucial anomaly subgraph detection task for ensuring urban security. Typically, identifying criminal activity in urban street networks (USNs) or urban transport networks (UTNs) is often impossible or limited by insufficient road surveillance, *e.g.*, in a city's commercial district, traffic monitoring may report normal flow while social media simultaneously reveals abnormal events such as robberies or sudden crowd gatherings. As shown in Fig. 1, emotional words (*e.g.*, nasty, hate, and complaint) frequently appear in social networks (SNs), and this particularly anomalous attribute is difficult to observe in USN. Similar situations also arise in public SNs and private forums. The private nature of some SNs conceals certain information, making it challenging to directly capture the anomalous attributes. In addition to the absence of anomalous attributes in the dataset, the lack of anomaly features may also result from anomalous attributes being obscured within the vast volume of big data and its high-dimensional nature. How can we achieve effective detection performance when anomaly features are lacking due to the absence of observable anomalous attributes?

¹In this article, we use the terms “network” and “graph” interchangeably for the structure denoted as $G = (V, E)$, where V and E denote the node and the edge sets, respectively.

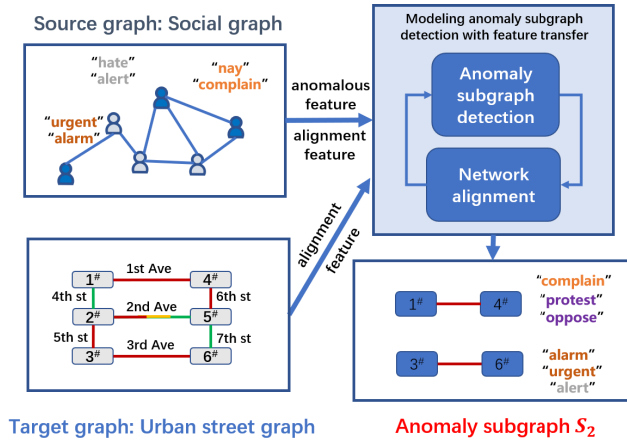


Fig. 1. Illustration of crime hotspot detection using the IASD framework. Crime hotspots in the urban street graph are identified by leveraging a social graph enriched with negative emotional attributes. The proposed IASD approach integrates anomaly subgraph detection with network alignment.

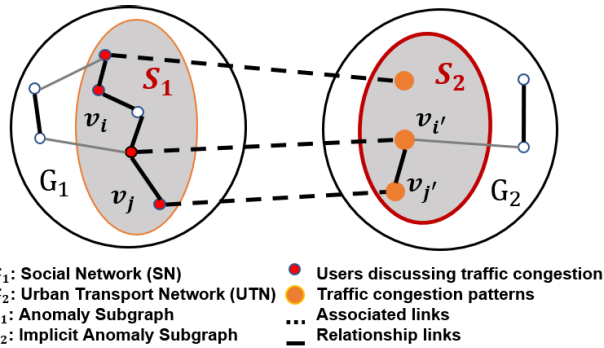


Fig. 2. Illustration of anomaly alignment: red circles v_i and v_j denote anomaly nodes, orange circles v'_i and v'_j denote implicit anomaly nodes, solid lines (v_i, v_j) and (v'_i, v'_j) are edges in graphs, and dash lines (v_i, v'_i) and (v_j, v'_j) are the alignment edges between two graphs.

In this article, we refer to the anomaly subgraph hidden within the target data that lacks anomaly features as the implicit anomaly subgraph (IAS). Hence, detecting anomaly subgraphs will face challenges due to the *lack of anomaly features within the target datasets*. Another challenge encountered by many existing anomaly subgraph detection algorithms is the *lack of interpretability*. Especially, single-attribute networks hinder anomaly cause identification without sufficient features, as seen in UTN data where sparse accident metadata impedes accurate congestion analysis. *Such discrepancies motivate the development of anomaly feature transfer learning algorithms that leverage social signals to complement traffic/street monitoring data and enable early detection of potential risks.*

To address the two aforementioned challenges, our approach incorporates associated auxiliary data to enable the identification of IASs and underlying anomaly causal chains within the target data. An anomalous event often exhibits correlations across multiple domain-specific datasets. This correlation is termed as “Anomaly Alignment” in this article. Another instance is that unusual opinions about traffic congestion reported on SNs and congestion patterns recorded by UTNs often coincide at specific street locations or geographic coordinates, as illustrated in Fig. 2. Consequently, we construct the causal chain underlying a traffic congestion event by

analyzing both opinion diffusion subgraphs and congested subgraphs. Inspired by transfer learning to address the challenge of data sparsity and sample limitations across domains and tasks, this article proposes an innovative anomaly feature transfer learning approach for detecting IAS through *Anomaly Alignment across multidimensional attributed networks*.

Our approach works on a multilayer graph structure $N = (G_1, G_2, M)$ to tackle the aforementioned challenges. G_1 is the source graph associated with the target graph G_2 . G_1 contains certain observable anomalous attributes, which make it easier to capture anomaly features and detect the anomaly subgraph S_1 within it. In contrast, the target graph G_2 lacks explicit anomalous attributes and includes the IAS, referred to as S_2 . Let M denote the association relationships between G_1 and G_2 . These association relationships are modeled within a two-layer graph N , subject to specific alignment constraints. Although this article focuses on two-layer graphs, the proposed method can be extended to multilayer graphs. A multilayer graph is an effective model for representing multiple interrelated source graphs originating from diverse domains. Each layer represents a unique type of interaction between nodes, and different layers consist of distinct sets of nodes and types of links. Additionally, alignment links exist between layers to represent the association relationships across different graphs [3]. Our approach postulates that the anomaly features present in the source graph G_1 affect the target graph G_2 through alignment links M between G_1 and G_2 . Hence, the IAS S_2 in G_2 is possibly detected by correlating this affection from G_1 . Fig. 1 demonstrates our approach for detecting urban crime hotspots in social and USNs. Our main contributions can be summarized as follows.

- 1) This article proposes a pioneering approach (IASD), designed for the specific challenge of detecting IASs S_2 under “Anomaly Alignment” within multidimensional attribute networks.
- 2) IASD novelty interprets the anomaly subgraph S_2 by fusing aligned attributes from the explicit anomaly subgraph S_1 in G_1 , facilitating the detection of causal chains within anomalies.
- 3) Extensive experiments are conducted on five real-world datasets, showcasing IASD’s significant performance and robustness. Additionally, IASD is applied to address four practical application tasks.

The remainder of this article is structured as follows. Section II provides an overview of relevant existing methods, while Sections III–IV define the “anomaly alignment” problem and introduce IASD in detail. Sections V–A–V encompass five real datasets and offer a comprehensive evaluation of IASD. Four case studies are presented in Section V–E, and finally, Section VI summarizes the primary findings of this research.

II. RELATED WORKS

Table I summarizes representative graph anomaly detection (GAD) methods from recent years and compares them with our approach across six aspects. These methods were comprehensively compared in terms of anomaly types, consideration of structural anomalies, capability to handle few-attribute anomalies, suitability for multiple graphs, interpretability, and reliance on graph neural networks (GNNs).

TABLE I
COMPARISON BETWEEN OUR METHOD AND SOME EXISTING REPRESENTATIVE GAD MODELS

Method	Level of Anomalies	Structural Anomalies	Few Attribute Anomalies	Multiple Graphs	Interpretability	Based GNNs
SAMCL (2025) [4]	node	✓	✗	✗	✗	✓
RG-GLD (2024) [5]	edge	✓	✗	✓	✗	✓
HO-GAT (2023) [6]	node& Motif	✓	✗	✗	✗	✓
TSPSD (2018) [7]	subgraph	✗	✗	✗	✗	✗
NPHGS (2014) [8]	subgraph	✗	✗	✗	✗	✗
ADOMS (2017) [9]	node	✓	✗	✓	✗	✗
ASDHC (2024) [10], ASGAE (2022) [11]	subgraph	✓	✗	✗	✓	✓
A3MAN (2022) [12]	subgraph	✓	✗	✓	✓	✓
ASDFT (2020) [13], AAAN (2022) [14]	subgraph	✗	✓	✓	✓	✗
IASD	subgraph	✓	✓	✓	✓	✓

A. Anomaly Subgraph Detection

Traditional anomaly subgraph detection methods typically extract features from graph-structured data and identify anomalies via statistical and graph search techniques [1], [15]. These nondeep learning approaches handle static and dynamic graphs, including single, multiple, and multilayered graphs [7], [8], [9], [16], [17], but face challenges with large-scale graphs and high-dimensional node or edge attributes.

To address these challenges, recent approaches leverage deep learning [2], [18], [19]. However, due to inefficient subgraph representation learning, most focus on detecting anomalous nodes rather than subgraphs [4], [5], [20], [21], [22], [23]. For instance, Hu et al. [4] propose subgraph-aligned contrastive learning using multiview augmented subgraphs to detect anomalous nodes. Liu et al. [23] detect anomalies by evaluating data similarity through representative neighbors. Zhou et al. [5] propose a lightweight IoT anomaly detection model combining GNN and knowledge distillation with global-local feature learning for secure, efficient, and low-overhead communication.

Some deep learning methods target anomaly subgraphs by learning distinctive features. Huang et al. [6] propose a hybrid-order graph attention (GAT) network to detect abnormal nodes and motif instances. Zhang and Zhao [11] evaluate subgraph anomaly scores in a residual graph built from reconstructed and original graphs using a location-aware graph autoencoder. Sun et al. [10] introduce ASD-HC, which combines GCN-based contrastive learning with Nonparametric Graph Scan statistics to assess anomalous subgraphs.

In summary, while these methods learn anomaly features from data, most do not target IASs.

B. Anomaly Detection on Multiple Networks

Recently, some works study anomaly detection across multiple networks. Bindu et al. [9] detect anomalous users by comparing topological similarities across layers. Sun et al. [14] propose AAAN, using ADMM to find maximum connected anomaly subgraphs across networks. Wu et al. [12] apply deep learning and network alignment to detect shared anomalies in traffic networks. However, there are also works that can be easily confused with anomaly subgraph detection research, such as anomaly or object detection across multiple views [24] and domains [25]. These approaches do not take into account the network structure of the data.

Other GNN-based GAD methods, like SAMCL [4], focus on rich anomaly features and complex topologies, while multilayer methods such as A3MAN [12] emphasize interpretability. Our approach is more general, detecting IASs in two-layer, multidimensional networks, handling missing

features, leveraging GNNs for topological anomalies, enabling anomaly feature transfer across domains like pandemic and traffic networks, and providing good interpretability.

C. Transfer Learning

Existing transfer learning methods address challenges such as data scarcity, domain shift, task differences, computational constraints, negative transfer, and heterogeneous transfer. By leveraging knowledge from a source domain—e.g., model parameters or feature representations—they mitigate limited labeled data, distribution discrepancies, and task mismatches in the target domain, improving performance and generalization [26], [27], [28], [29], [30], [31], [32], [33], [34].

Transfer learning can be categorized into four subsettings: *Instance-transfer*, *Feature-representation-transfer*, *Parameter-transfer*, and *Relational-knowledge-transfer* [35]. In instance-transfer, Askarizadeh et al. [26] propose an optimization framework to improve learning under resource constraints while reducing negative transfer, and Liang et al. [36] extend instance-transfer to large-scale SNs. In feature-representation transfer, Li et al. [27] introduce a cross-channel specific-mutual feature transfer network for EEG decoding, Gao et al. [29] propose a multitask model to extract shared features across tasks, Lu et al. [30] tackle cross-modality person re-identification by learning both common and modality-specific features, and Liang et al. [36] proposed an instance-transfer method for large-scale SNs.

In relational-knowledge transfer, Tang et al. [28] leverage social theories for transfer across heterogeneous networks, while Bi et al. [32] develop Bridged-GNN, enhancing transfer via a Bridged-Graph structure. Cui et al. [31] apply knowledge transfer in human-machine dialog systems to improve generalization. Beyond traditional transfer learning, Li et al. [33], [34] explore domain adaptation for robust object detection across domains and unseen categories.

In summary, these studies highlight the diverse methods and applications of transfer learning, providing a foundation for further research in anomaly detection and knowledge transfer in complex structured domains.

III. PROBLEM FORMULATION

Consider a two-layer graph $N = (G_1, G_2, M)$, where $G_1 = (V_1, E_1, X_{G_1})$ is the source graph (layer), $G_2 = (V_2, E_2, X_{G_2})$ is the target graph (layer), and $M \subseteq V_1 \times V_2$ represents the set of association relationships (i.e., alignment edges) between nodes in G_1 and G_2 . Specifically, $V_1 = \{v_{11}, v_{12}, \dots, v_{1n}\}$ and $V_2 = \{v_{21}, v_{22}, \dots, v_{2m}\}$ are the node sets, with $E_1 \subseteq V_1 \times V_1$ and $E_2 \subseteq V_2 \times V_2$ as their corresponding edge sets. The attribute matrices are given by $X_{G_1} = \{x_{G_1}(v_i) | v_i \in V_1\} \in \mathbb{R}^{n \times d_1}$ and

$X_{G_2} = \{x_{G_2}(v_j) | v_j \in V_2\} \in \mathbb{R}^{m \times d_2}$, where $x_{G_1}(v_i) \in \mathbb{R}^{d_1}$ and $x_{G_2}(v_j) \in \mathbb{R}^{d_2}$ denote the multidimensional attribute vectors of nodes in G_1 and G_2 , respectively. The alignment edges in M model cross-layer associations, enabling anomaly feature transfer from G_1 to G_2 . Accordingly, we formalize the following foundational definitions.

Definition 1 (Anomalous Node): Given a graph $G = (V, E, X)$, let $p(X)$ denote the probability distribution of the node features in G . For a node $v \in V$, if the probability of its feature vector satisfies $p(x_G(v)) \leq \alpha$, where $\alpha \in \mathbb{R} \geq 0$ is a predefined significant anomaly threshold, then node v is considered an anomalous node of G .

Definition 2 (Anomaly Subgraph): Given a subgraph $S = (V_S, E_S, X_S)$, if an anomalous node $v \in V$ exists or X_S comprises the anomaly features, then the subgraph S is considered an anomaly subgraph.

We denote $S_1 = (V_{S_1}, E_{S_1}, X_{S_1})$ and $S_2 = (V_{S_2}, E_{S_2}, X_{S_2})$ as the anomaly subgraphs of G_1 and G_2 , respectively. When X_{G_1} contains anomaly attributes while X_{G_2} lacks such attributes, S_1 can be readily detected from the source graph G_1 through the anomaly attributes in X_{G_1} . However, detecting S_2 directly from graph G_2 proves challenging. If G_1 aligns with G_2 under certain association relationship constraints, then the node anomaly attributes of G_1 can be projected onto the corresponding aligned nodes in G_2 . This naturally raises a critical question: Can the IAS S_2 be effectively detected from G_2 through the joint utilization of attributes X_{G_1} and X_{G_2} ?

Definition 3 (Feature Transfer Learning): Given $G_1 = (V_1, E_1, X_{G_1})$ is the source graph and $G_2 = (V_2, E_2, X_{G_2})$ is the target graph, with alignment edges $M \subseteq V_1 \times V_2$, the general feature transfer learning model $f_{\text{tran}} : X_{G_1} \rightarrow X_{G_2}$ is given as follows:

$$X_{G_2} \leftarrow X_{G_2} + f_{\text{tran}}(X_{G_1}, M) = X_{G_2} + \Phi(\mathbf{M}X_{G_1}) \quad (1)$$

where $\mathbf{M}^{m \times n}$ is the matrix form of the alignment edge set M , Φ is a feature projection function.

Proposition 1 (Anomaly Alignment): Given a two-layer graph $N = (G_1, G_2, M)$, where $G_1 = (V_1, E_1, X_{G_1})$ is the source graph and $G_2 = (V_2, E_2, X_{G_2})$ is the target graph, with alignment edges $M \subseteq V_1 \times V_2$. A connected anomaly subgraph $S_1(V_{S_1}, E_{S_1}, X_{S_1})$ in G_1 contains an anomalous node v_i . If $(v_i, v_j) \in E_{S_1}$ and $(v'_i, v'_j) \in E_2$ exist with $(v_i, v'_i) \in M$ and $(v_j, v'_j) \in M$, then according to our feature transfer learning model $f_{\text{tran}} : X_{S_1} \rightarrow X_{S_2}$, a nonempty aligned anomaly subgraph S_2 exists in G_2 .

Proof: The detailed proof is provided in the appendix. $\square$

Definition 4 (IASD): IAS detection (IASD) refers to detecting the maximum anomaly subgraph within the target network, which is devoid of anomalous attributes.

Under the constraint of the Anomaly Alignment, the objective function of IASD is defined as the following constrained optimization problem:

$$\begin{aligned} & (\hat{S}_1, \hat{S}_2, \hat{M}) \\ & = \arg \max_{S_1 \subseteq G_1, S_2 \subseteq G_2, M} F_\alpha(S_1) + Q(S_1, S_2, M) \\ & \text{s.t. } S_1 \text{ is connected, } M \in \{0, 1\}^{m \times n}, |V_{S_2}| \leq c, \alpha \in \mathbb{P} \end{aligned} \quad (2)$$

where α is a hyperparameter which represents the anomalous significant level, the set $\mathbb{P} = \{p(x_{G_1}(v_i)) \mid p(x_{G_1}(v_i)) \leq \alpha_{\max}, v_i \in V_1\} \cup \{\alpha_{\max}\}$ comprises the probabilities of the significant anomalies of G_1 , with $\alpha_{\max} = 0.15$ in this article.

Here, $c \leq m$ is a constant representing the maximum number of nodes in S_2 . The function $F_\alpha(S_1)$ evaluates the anomaly score for the subgraph S_1 . Let $Q(S_1, S_2, M)$ be the feature transfer learning model based on network alignment principles [37], which scales the anomalous significance of S_1 transferred to S_2 . In our article, the similarity of S_1 and S_2 is represented by the alignment matrix $\mathbf{M} \in \mathbb{R}^{m \times n}$. Let $\mathbf{M}(j, i) = 1$ if node $v_i \in V_{S_1}$ aligns with the node $v_j \in V_{S_2}$, otherwise $\mathbf{M}(j, i) = 0$.

Intuitively, IASD problem (2) has the following three properties.

- 1) $P1$: Function $F_\alpha(S_1)$ increases monotonically with $N_\alpha(S_1)$, i.e., the number of anomaly nodes of S_1 .
- 2) $P2$: Function $F_\alpha(S_1)$ decreases monotonically with $\bar{N}_\alpha(S_1)$, i.e., the number of normal nodes in S_1 .
- 3) $P3$: Function $Q(S_1, S_2, M)$ increases monotonically with the number of alignment edges between S_1 and S_2 .

$P1$ and $P2$ guarantee that the optimal anomaly subgraph $\hat{S}_1$ detected by maximizing the value of function $F_\alpha(S_1)$ [7], and $P3$ guarantees that the optimal anomaly subgraph $\hat{S}_1$ contains more nodes of $\hat{S}_2$ that are aligned with the anomaly nodes of $\hat{S}_1$. These three properties guarantee that the induced anomaly subgraph $\hat{S}_2$ is the optimal result. The corresponding formal proof is provided in Section IV, and additional detailed proofs are included in Appendix. Note that in this article, we do not impose any connectivity constraints on S_2 .

IV. METHODOLOGY

A. Feature Extraction Preprocessing

With the widespread utilization of large-scale data, high-dimensional attributes are widely adopted to characterize nodes and edges in large-scale graphs. Consequently, anomaly subgraph detection algorithms will face the challenge of the curse of dimensionality in high-dimension datasets, making it difficult to observe anomalous attributes. In the process of extracting anomalous features for each node in graphs, our approach incorporates the GNN model (e.g., GCN, GAT, etc.) to convert nodes' high-dimensional attributes into a low-dimensional representation of anomalous features. In this article, we investigate the GAT network model to extract node features for both the source graph G_1 and the target graph G_2 . Now, let us delve into the detailed procedure using G_1 as an example. First, the initial nodes' attributes of G_1 are represented as $X_{G_1} = \{x_{G_1}(v_i) | v_i \in V_1\}$, where $x_{G_1}(v_i) \in \mathbb{R}^{1 \times d_1}$. The normalized attention coefficient from node v_j to node v_i is calculated by the following equation:

$$a_{ij} = \frac{\exp(\text{LeakyReLU}(\mathbf{a}^T [Wx_{G_1}(v_i) \parallel Wx_{G_1}(v_j)]))}{\sum_{k \in N_i} \exp(\text{LeakyReLU}(\mathbf{a}^T [Wx_{G_1}(v_i) \parallel Wx_{G_1}(v_k)]))} \quad (3)$$

where N_i denotes the neighbor set of node v_i in graph G_1 , $W \in \mathbb{R}^{d'_1 \times d_1}$ represents the weight matrix used for linear transformation applied to each node, d'_1 is the dimension of the output feature for each node. $\mathbf{a} \in \mathbb{R}^{2d'_1}$ is the weight vector, $\mathbf{a}^T$ denotes the transposition of vector $\mathbf{a}$. $\parallel$ denotes the concatenation operation. LeakyReLU is a nonlinear activation function characterized by a negative input slope of 0.2. Then, the final output feature of node v_i is computed through a linear combination of the features as follows:

$$h_1(v_i) = \sigma \left(\sum_{v_j \in N_i} a_{ij} Wx_{G_1}(v_j) \right). \quad (4)$$

To obtain stable learning features, we employ the aggregation of a multihead GAT mechanism, which utilizes K -independent attention mechanisms and applies averaging computation to the final output features of each node, as shown in the following equation:

$$h_1(v_i) = \sigma \left(\frac{1}{K} \sum_{k=1}^K \sum_{v_j \in N_i} a_{ij}^k W^k x_{G_1}(v_j) \right) \quad (5)$$

where K is the number of independent attention mechanisms, and σ represents the nonlinear activation function.

Similarly, the feature X_{G_2} in G_2 can also be generated following the aforementioned process. Thus, $\mathbf{x}_1 \in \mathbb{R}^{n \times 1}$ and $\mathbf{x}_2 \in \mathbb{R}^{m \times 1}$ are feature vectors of G_1 and G_2 , calculated as shown in the following equation:

$$\begin{aligned} \mathbf{x}_1(i) &= \|h_1(v_i)\|_2, \quad \text{s.t. } v_i \in G_1 \\ \mathbf{x}_2(i) &= \|h_2(v_i)\|_2, \quad \text{s.t. } v_i \in G_2. \end{aligned} \quad (6)$$

B. Anomaly Alignment Model

Fig. 2 depicts the *Anomaly Alignment* proposition. Now, let us elaborate on our proposed approach. Anomaly alignment comprises two main stages.

1) *Detecting Anomaly Subgraph S_1 in G_1* : In this stage, we introduce a NPGS-based method (TSPSP [7]) to detect the optimal anomaly subgraph. Specifically, NPGS-based methods calculate the anomaly score function $F_\alpha(S_1)$ for each subgraph S_1 in G_1 through NPGS statistics to detect the optimal anomaly subgraph S_1 of G_1 . Therefore, the general formulation of the anomaly score function $F_\alpha(S_1)$ in our problem (2) is defined as follows:

$$F_\alpha(S_1) = \varphi(\alpha, N_\alpha(S_1), N(S_1)) \quad (7)$$

where α is the anomalous significant level, $N_\alpha(S_1)$ is the number of anomaly nodes in S_1 , and $N(S_1)$ is the total number of nodes in S_1 . The function $\varphi : V_{S_1} \rightarrow \mathbb{R}$ is the NPGS function used to evaluate the anomalous extent for subgraph S_1 . In this article, We explore two NPGS statistics within TSPSD: Berk-Jones (BJs) [38], as defined in (8), and higher criticism (HC) [39]

$$\varphi_{BJ}(\alpha, N_\alpha(S), N(S)) = N(S) \times KL \left(\frac{N_\alpha(S)}{N(S)}, \alpha \right) \quad (8)$$

where KL is Kullback–Liebler divergence between the observed and expected proportions, with values less than α

$$KL(a, b) = \begin{cases} a \log \left(\frac{a}{b} \right) + (1-a) \log \left(\frac{1-a}{1-b} \right), & \text{if } a \geq b \\ 0, & \text{if } a < b \end{cases} \quad (9)$$

where $a, b > 0$.

Lemma 1 [7]: Given $\bar{N}_\alpha(S_1) = N(S_1) - N_\alpha(S_1)$, the optimal subgraph of NPGS problem with the K -budget node-weighted prize-collection Steiner tree is defined as

$$S_1^{\alpha, K} = \arg \max_{S_1 \subseteq G_1, S_1 \text{ is connected}} N_\alpha(S_1), \quad \text{s.t. } \bar{N}_\alpha(S_1) \leq K \quad (10)$$

where α is the anomaly significance level, typically set as $\alpha \leq 0.15$ in this article. $K \leq n$ is the number of normal nodes in S_1 .

Lemma 2 [7]: Given $S_1^{\alpha, K}$, ($K = 0, \dots, n$) is the optimal solution of the K -budget node-weighted prize-collection Steiner Tree from (10), the NPGS problem

$$\hat{S}_1 = \arg \max_{S_1 \subseteq G_1, S_1 \text{ is connected}} F_\alpha(S_1)$$

is equivalent to the following optimal problem:

$$\hat{S}_1 = \arg \max_{\alpha \in \mathbb{P}, S_1 \in \{S_1^{\alpha, 0}, \dots, S_1^{\alpha, n}\}} F_\alpha(S_1). \quad (11)$$

In summary, TSPSD decomposes the NPGS-based anomaly subgraph detection problem into a series of K -budget subgraph detection subproblems. With the constraints of lemma 5 and 6, we guarantee that the optimal solution $\hat{S}_1$ in (11) is the optimal anomaly subgraph detected from G_1 .

2) *Anomaly Transferring From G_1 to G_2* : Inspired by transfer learning the biological knowledge among protein interaction networks based on the network alignment [40], [41], [42], our algorithm calculates the S_2 's anomaly features by transferring the S_1 's anomaly features through the network alignment. Our algorithm transfers the anomaly features of S_1 to infer the anomaly features of S_2 , while simultaneously ensuring optimal structural alignment between the associated graphs G_1 and G_2 . Our anomaly transferring model is formalized as the following convex combination form:

$$Q(S_1, S_2, M) = \text{ICS}(S_1, S_2, M) + \text{FT}(S_1, S_2, M). \quad (12)$$

Here, $\text{ICS}(S_1, S_2, M)$ denotes the topological similarity function based on the conserved aligned edges [37], [43]

$$\begin{aligned} \text{ICS}(S_1, S_2, M) &= \frac{\sum M(v'_i, v_i) M(v'_j, v_j) A_1(v_i, v_j) A_2(v'_i, v'_j)}{\sum_{v'_i, v'_j} A_2(v'_i, v'_j)} \end{aligned} \quad (13)$$

where A_1 and A_2 are the adjacency matrices of G_1 and G_2 , $v_i, v_j \in V_{S_1}$, $v'_i, v'_j \in V_{S_2}$, $\mathbb{I}(\cdot)$ is the indicator function.

$\text{FT}(S_1, S_2, M)$ represents the induced subgraph features that combine the inherent features of S_2 and the multidimensional features transferred from S_1 . To enhance significant anomaly features, we modify our feature transfer learning model (1) into the following general function:

$$\text{FT}(S_1, S_2, M) = X_{S_2} + \Phi(M(1 - X_{S_1})). \quad (14)$$

Therefore, the objective function of our anomaly transfer model is defined below

$$\arg \max_{\alpha \in \mathbb{P}, S_1, S_2, M} Q(S_1, S_2, M), \quad \text{s.t. } \bar{N}_\alpha(S_1) \leq K, |S_2| \leq c \quad (15)$$

where $K \leq n$ and $c \leq m$ are both hyperparameters.

Given the constraint of hyperparameters, achieving the optimal value of $Q(S_1, S_2, M)$ requires simultaneously maximizing both the ICS and FT functions, as shown in (12). Function $Q(S_1, S_2, M)$ maximizes the alignment quality between S_1 and S_2 , while highlighting the significant induced anomaly features. Furthermore, $F_\alpha(S_1)$ in (7) increases monotonically with the number of anomaly nodes $N_\alpha(S_1)$ and decreases with the number of normal nodes $\bar{N}_\alpha(S_1)$. Based on Lemma 6, the optimal anomaly subgraph S_1 exists in the limited graph G_1 . Hence, the significant anomaly subgraph $\hat{S}_1$ can be directly identified from G_1 using the method in [7]. Subsequently, ICS function facilitates the identification of the most structurally similar subgraph alignment pairs, while FT function ensures

Algorithm 1 IASD With Multidimensional Feature Transfer

Input: $N = (G_1, G_2, H)$, $G_1 = (V_1, E_1, X_{G_1})$, $G_2 = (V_2, E_2, X_{G_2})$, Number of S_2 ' nodes c , $\alpha_{\max}$

Output: The optimal implicit anomaly subgraph $\hat{S}_2$

- 1 Initialization: $M^0 \leftarrow H$, $S_1^0 \leftarrow \phi$, $S_2^0 \leftarrow \phi$;
- 2 Extracting anomaly feature vectors in graph G_1 via GAT: $\mathbf{x}_1 \leftarrow G_1$;
- 3 Extracting normal feature vectors in graph G_2 via GAT: $\mathbf{x}_2 \leftarrow G_2$;
- 4 Calculating the probability of features for nodes in G_1 : $\{p(v_i)\} \leftarrow \mathbf{x}_1$
- 5 **for** $\alpha \in \mathbb{P} = \{p(v_i) | p(v_i) \leq \alpha_{\max}\} \cup \{\alpha_{\max}\}$ **do**
- 6 **for** $K \leq n$ **do**
- 7 $S_1^K \leftarrow \max_{S_1} N_\alpha(S_1)$, s.t. $\bar{N}_\alpha(S_1) \leq K$, S_1 is connected;
- 8 $S_2^K \leftarrow \max_{S_2} Q(S_1^K, S_2, M^{K-1})$, s.t. $|V_{S_2}| \leq c$;
- 9 $M^K \leftarrow \arg \max Q(S_1^K, S_2^K, M^{K-1})$;
- 10 **end for**
- 11 $(S_1^\alpha, S_2^\alpha, M^\alpha) \leftarrow \max_K (F_\alpha(S_1^K) + Q(S_1^K, S_2^K, M^K))$;
- 12 **end for**
- 13 $(\hat{S}_1, \hat{S}_2, \hat{M}) \leftarrow \max_\alpha (F_\alpha(S_1^\alpha) + Q(S_1^\alpha, S_2^\alpha, M^\alpha))$;
- 14 **return** $\hat{S}_2$;

the effective transfer of significant anomaly features from the source graph to the target graph, resulting in strongly induced anomaly patterns. Therefore, $Q(S_1, S_2, M)$ is monotonically increasing with the number of alignment edges ($|M|$) and the number of anomaly nodes in S_1 , i.e., $N_\alpha(S_1)$. Our algorithm IASD satisfies properties P1, P2, and P3 described in Section III. The validity of these conclusions is confirmed by parameter-sensitivity experiments, the details of which are provided in the appendix.

C. IASD With Multidimensional Feature Transfer Algorithm

IASD is summarized below, as shown in Algorithm 1.

- 1) Fusing the multidimensional attributes into the feature vector $\mathbf{x}_1$.
- 2) Optimizing the IASD problem (2) via dynamic programming procedure, and deriving the optimal anomaly subgraph $\hat{S}_2$.
 - a) Detecting the maximum anomaly subgraph S_1 by calculating Function $F_\alpha(S_1)$, based on the anomaly feature $\mathbf{x}_1$.
 - b) Calculating the optimal subgraph S_2 by maximizing Function Q .
 - c) Updating the alignment matrix $\mathbf{M}^{m \times n}$ based on the new S_1 and S_2 .

Lemma 3: Optimal anomaly S_2 with S_1 and $\mathbf{M}$ fixed. Under the condition that the anomaly subgraph S_1 and the alignment matrix $\mathbf{M}$ are fixed, if S_1 and S_2 satisfy Anomaly Alignment (Proposition 1), then the maximum solution $\hat{S}_2$ of Function Q is defined as follows:

$$\hat{S}_2 = \lambda \mathbf{M} [\mathbf{s}_1 \odot (\mathbf{1} - \mathbf{x}_1)] + (\lambda \mathbf{x}_2^T + 1) \mathbf{M} \mathbf{s}_1 \quad (16)$$

where λ is a regularization parameter.

Proof: The detailed proof is provided in the appendix. $\square$

Lemma 4: Optimal alignment matrix $\mathbf{M}$ with S_1 and S_2 fixed. Given two input subgraphs S_1 and S_2 , then Function Q in (15) w.r.t. $\mathbf{M}$ is equivalent to the optimal problem

$$\hat{\mathbf{M}} = \arg \max_{\mathbf{M} \in [0,1]^{m \times n}} \eta [Q(\mathbf{M})] - (1 - \eta) \|\mathbf{M} - \mathbf{H}\|_F^2 \quad (17)$$

where η is a regularization parameter, $\mathbf{H}$ is the prior alignment matrix, and $\|\cdot\|_F$ denotes the Frobenius norm.

Proof: The detailed proof is provided in the appendix. $\square$

Theorem 1: Optimal anomaly subgraphs S_1, S_2 in Graphs G_1, G_2 . If subgraphs S_1 and S_2 satisfy the anomaly alignment condition, then in the IASD problem [(2)], an optimal solution exists and is defined as follows:

$$(\hat{S}_1, \hat{S}_2) = \arg \max_{\alpha \in \mathbb{P}} F_\alpha(S_1) + Q(S_1, S_2, M). \quad (18)$$

Proof: The detailed proof is provided in the appendix. $\square$

a) **Complexity analysis:** The time complexity of IASD primarily relies on the iterations involving S_1^K, S_2^K , and M^K . Detecting the anomaly subgraph S_1^K involves solving a 0-1 multiple-choice knapsack (MCK) Problem, which can be addressed using dynamic programming. Its worst case time complexity is $O(|\mathbb{P}| \cdot n^3)$ for a graph with n nodes [7]. To optimize $\mathbf{M}$ efficiently, we exploit the Kronecker product property, yielding a time complexity of $O(|\mathbb{P}| \cdot mn^2)$ for this step. Additionally, the time complexity of S_2^K is also $O(|\mathbb{P}| \cdot mn^2)$. Hence, the time complexity of our algorithm is $O(|\mathbb{P}| \cdot (n^3 + mn^2))$. When $m = O(n)$, this reduces to $O(|\mathbb{P}| \cdot n^3)$, which is consistent with our empirical observation that the running time is dominated by the n^3 term. The evaluation results are presented in the appendix.

V. EXPERIMENTS**A. Dataset Description**

Fig. 3 shows four multilayer graphs from five real datasets (details in the appendix).

- 1) *Weibo Dataset* contains 17.26 million geo-tagged Weibo users from Tianjin (December 1, 2011–December 31, 2012), collected from www.weibo.com. Applying SentiCircle [44], 99 872 negative tweets are filtered to build the source graph G_1 (*Negative Friendship Graph*) with 11 225 users and 13 405 links. The target graph G_2 (*Neighborhood Graph*) has 137 919 nodes and 6 396 462 edges, where nodes are neighborhoods (areas with a diameter of 300 m) and edges indicate a user tweeting across them.
- 2) *TYC Dataset* comprises 1.67 million corporations, 1.0 million investors, and 2.66 million investment relationships from www.tianyanan.cha.com (January 1, 2000–December 31, 2018). The source graph G_1 contains 864 microcredit loan companies and 1814 relationships, while the target graph G_2 has 4785 shareholders and 35 892 co-investment links. Companies labeled as risky on the website were collected as ground truth.
- 3) *B&C Datasets:* *B dataset* comprises 4.8 million population movement records among 441 Chinese cities from www.Baidu.com (January 1–February 29, 2020), where edges represent the top 100 movements between cities, forming G_1 of B&C. *C dataset* includes 24 787 daily confirmed COVID-19 cases and 7686 patient trajectories (December 1, 2019–March 10, 2020) across 313 Chinese cities. G_2 of B&C is constructed as a plain graph with 313 nodes and 7686 edges, using only the topology and treating COVID-19 data as ground truth.
- 4) *Water Pollution (WP) Dataset* was collected from a real Water Sensor Network across six time slices, comprising 12 527 sensors (nodes) and 14 831 edges monitoring

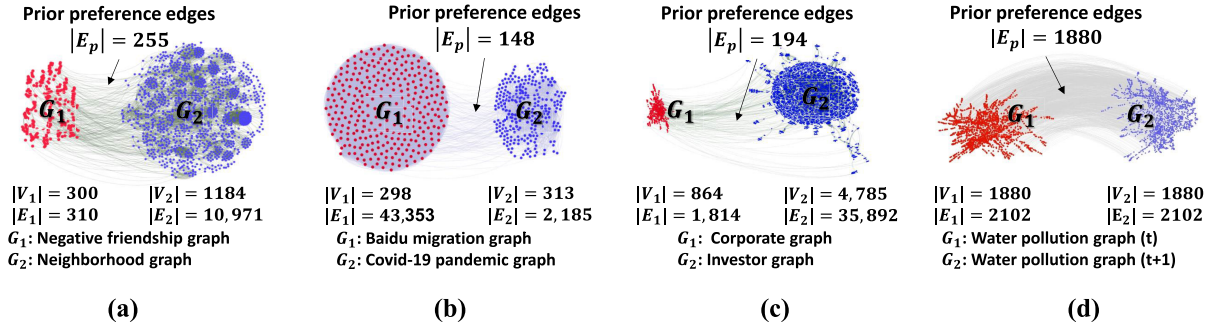


Fig. 3. Description of multilayer graphs in our case study experiments. In each subillustration, the source graph G_1 exhibits intrinsic anomalous characteristics, where explicitly annotated observable anomalies are encoded in red, while anomaly alignment between G_1 and target graph G_2 are delineated through gray directional arrows symbolizing cross-domain alignment mechanisms. G_2 obtain induced anomalous features from G_1 through transfer learning, enabling G_2 to detect the IAS (depicted in blue in the subillustration). (a) Urban and rural area crime hotspots detection. (b) Anomaly event detection: epidemic breakout. (c) Risky investors detection. (d) Water Pollution Prediction.

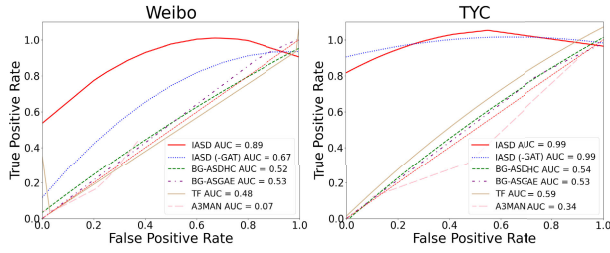


Fig. 4. ROC curves for evaluations of the IASD on the Weibo and TYC datasets.

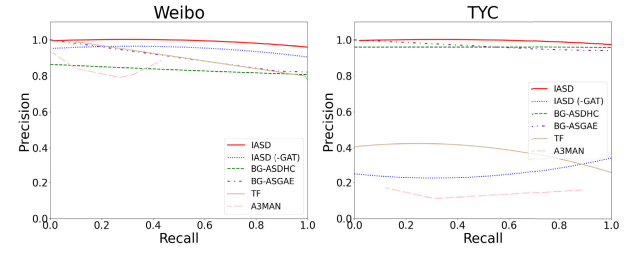


Fig. 5. Precision-Recall Curves for evaluations of the IASD on the Weibo and TYC datasets.

chemical pollutants in rivers. Sensors output 1' when contamination is detected, otherwise 0'. The source graph G_1 is built from the structure and attributes at time t , and the target graph G_2 from the structure at time $t + 1$.

B. Experiment Setup

All performance evaluations explore the BJ and HC statistics on four multilayer graphs, labeling algorithms with suffixes “(BJ)” or “(HC)”, e.g., IASD(BJ) and IASD(HC). As IASD(HC) results closely match IASD(BJ), they are provided in the appendix for brevity. For GAT layers, 2 heads are used for WP and 4 heads for TYC and Weibo, with attention weights initialized uniformly.

1) *Baselines*: We adopt five algorithms as our baselines.

- 1) *TF* is our designed baseline, which directly integrates the network alignment algorithm FINAL [40] with TSPSD and serves to detect an IAS.
- 2) *A3MAN* is a deep learning-based anomaly detection approach across multiple graphs [12].
- 3) *ASDHC* [10] employs GNNs and contrastive learning to detect anomaly subgraphs in an attributed network.
- 4) *BG-ASDHC* is our constructed baseline for detecting IASs, which consists of a GNN-based knowledge transfer learning algorithm *Bridged-GNN* [32] and a GNN-based anomaly subgraph detection algorithm *ASDHC*.
- 5) *BG-ASGAE* is our constructed baseline for detecting IASs, which integrates the *Bridged-GNN* [32] algorithm with a GNN-based anomaly subgraph detection algorithm *AS-GAE* [11].

2) *Evaluation Metrics*: Performance is evaluated by precision, recall, accuracy, false positive rate (FPR), and $F1$,

corresponding to true positives among detected positives, true detected positives among actual positives, correct detections overall, false detections, and combined precision-recall, respectively.

C. Experimental Result Analysis

1) *Performance Measures Analysis*: As shown in Table II, IASD achieves the highest scores across all metrics on the Weibo dataset, with improvements of 9% ~ 29% over competitive baselines. On the WP dataset, it outperforms four baselines in most metrics, except Recall, where it is 6% lower than TF(BJ). In the TYC dataset, IASD leads in Accuracy, Recall, and $F1$, while TSPSD(BJ) is 3% better in Precision and 8% lower in $FPRs$ (including FPR , $FPR-P$ and $FPR-D$). However, TSPSD(BJ) shows a 33% ~ 47% decrease in other metrics compared to IASD. Moreover, we demonstrate its superior performance by analyzing the correlation between TPR and FPR, as well as the correlation between precision and recall, as shown in Figs. 4 and 5. In summary, our approach outperforms nearly all the baseline methods.

2) *Robustness Evaluation*: To evaluate the robustness of our algorithm, we randomly selected a percentage of vertices and flipped their binary feature values. In addition, we make the noise ratio take values in the range [0%, 30%]. Compared to the baselines, IASD(BJ) has stable performance and achieves the best *Precision* and *FPR* under different noise rates, as shown in Fig. 6. It is worth noting that when the noise ratio is greater than 15, the precision shows an upward trend. Our analysis suggests this is because multiple abnormal subgraphs are adjacent to each other and are detected as one optimal abnormal subgraph. When the noise ratio exceeds a certain level, multiple original abnormal subgraphs connect with several “false anomaly nodes.” These false nodes are distributed in the adjacent areas of the smaller original

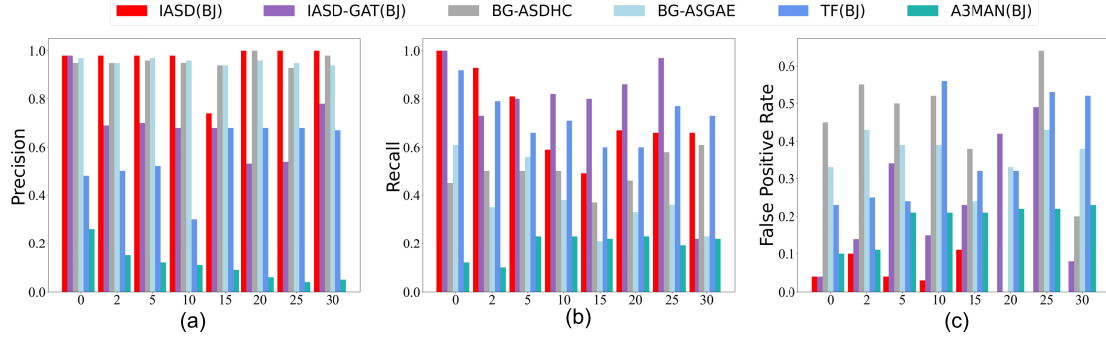


Fig. 6. Robustness evaluation on the TYC dataset under different noise ratios. (a) Precision. (b) Recall. (c) False positive rate.

TABLE II
COMPARISON BETWEEN IASD AND OTHER METHODS ON WP, WEIBO, AND TYC DATASET

Dataset	Method	Accuracy	Recall	F1	Precision	FPR	TPR-P	TPR-D	FPR-P	FPR-D
Weibo	TF(BJ)	0.77	0.77	0.81	0.91	0.15	0.77	0.77	0.27	0.29
	A3MAN(BJ)	0.42	0.39	0.47	0.76	0.42	0.40	0.33	0.39	0.50
	ASDHC	0.39	0.32	0.47	0.85	0.26	0.67	0.08	0.67	0.08
	BG-ASDHC	0.46	0.43	0.56	0.82	0.39	0.75	0.75	0.14	0.14
	BG-ASGAE	0.33	0.21	0.33	0.84	0.16	0.52	0.52	0.05	0.05
	IASD(BJ)	0.91	0.91	0.95	1.00	0.00	0.91	0.89	0.00	0.00
TYC	TF(BJ)	0.80	0.92	0.63	0.48	0.23	0.43	0.92	0.36	0.23
	A3MAN(BJ)	0.72	0.12	0.16	0.26	<u>0.10</u>	0.07	0.24	0.11	0.05
	ASDHC	0.24	0.21	0.34	0.95	0.18	0.82	0.01	0.82	0.01
	BG-ASDHC	0.45	0.45	0.61	0.95	0.45	0.95	0.95	0.04	<u>0.04</u>
	BG-ASGAE	0.44	0.61	0.45	<u>0.97</u>	0.33	0.44	0.44	0.33	0.33
	IASD(BJ)	0.90	<u>0.90</u>	0.93	0.97	0.08	<u>0.90</u>	<u>0.90</u>	<u>0.08</u>	0.08
WP	TF(BJ)	0.81	0.88	0.81	<u>1.00</u>	0.00	0.65	0.65	0.49	0.49
	A3MAN(BJ)	0.16	0.07	0.13	0.87	0.09	0.10	0.10	<u>0.01</u>	<u>0.01</u>
	ASDHC	0.41	0.34	0.48	0.79	0.33	0.65	0.12	0.65	0.12
	BG-ASDHC	0.27	0.20	0.33	0.89	0.20	0.66	0.66	0.03	0.03
	BG-ASGAE	0.25	0.20	0.32	0.82	0.32	<u>0.68</u>	<u>0.68</u>	0.19	0.19
	IASD(BJ)	0.83	<u>0.82</u>	0.90	1.00	0.00	0.78	0.78	0.00	0.00

abnormal subgraphs. This results in the detection of more “true anomaly nodes.” Fig. 6 shows the experimental results with the BJ statistic on the TYC dataset and similar results are derived with the HC statistic.

D. Ablation Study

1) *GAT Module Analysis*: In our approach, the GAT module embeds node attributes by sampling neighbor features. To assess its impact, we compare IASD with IASD-GAT (without GAT) using ROC (Fig. 4) and Precision-Recall curves (Fig. 5), where IASD achieves the highest AUC and the best Precision-Recall value. An additional ablation study (Fig. 6) evaluates robustness under varying noise levels: IASD attains higher Precision and lower FPR than IASD-GAT across noise settings, with better Recall when noise is below 10. However, as noise increases, GAT aggregation reduces Recall by making anomalous nodes harder to distinguish.

2) *Transferability Analysis*: The transferability of IASD depends primarily on the functions $Q(S_1, S_2, M)$ and $FT(S_1, S_2, M)$. To validate this, we compare IASD with ASDHC (Table II), which detects anomaly subgraphs directly on the feature-deficient G_2 . The poor performance of the ASDHC method proves the transferability of our method. We also applied TSPSD directly to G_2 , but it did not detect anomaly subgraphs, further demonstrating the effectiveness of our framework.

TABLE III
ABLATION STUDY ON TYC

Method	Precision	Recall	F1	Accuracy	FPR
IASD(BJ)	0.97	0.90	0.93	0.90	0.08
IASD-GAT(BJ)	0.97	0.84	0.90	0.86	0.08
ASDHC	0.95	0.21	0.34	0.24	0.18

In summary, as shown in Table III, removing the GAT or Transferability components leads to suboptimal performance, demonstrating the importance of each component.

E. Case Study

1) *Urban Crime Hotspot Detection*: We hypothesize that the aggregation of negative emotions among residents increases the likelihood of abnormal events in their neighborhoods. To test this, we detected urban crime hotspots by aligning the “Negative Friend Graph” G_1 with the “Neighborhood Graph” G_2 and transferring auxiliary anomaly features. Daily empirical p -values for user attributes (December 1, 2011–December 1, 2012) yielded 397 anomaly subgraphs corresponding to neighborhoods. These hotspots were evaluated against the Urban Area Security Incident Report issued by a third-party authority. Fig. 7 showed the optimal subgraph S_2 on August 25, 2012, which contained eight crime hotspots, all confirmed by ground truth.

2) *Risky Investors Detection*: This case is to identify potentially risky investors by observing unusual behaviors of specific companies. G_1 with 864 microcredit loan companies

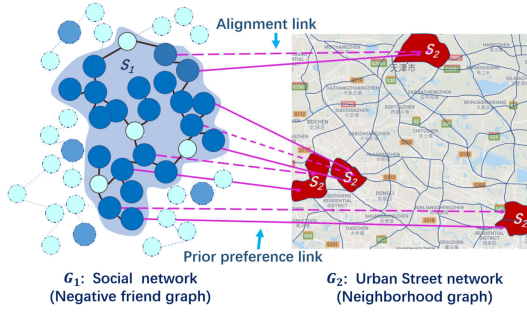


Fig. 7. Illustration of urban crime hotspot detection, G_1 consists of negative friends, G_2 represents the neighborhood graph. On August 25, 2012, IASD identified the optimal anomaly subgraph in G_1 with 21 anomaly users (dark blue circles) and detected 8 crime hotspots in G_2 (red areas). The pink solid lines denote prior preference links, while the dashed lines indicate alignment links. All hotspots are confirmed by the urban area security incident report from a third-party authority.

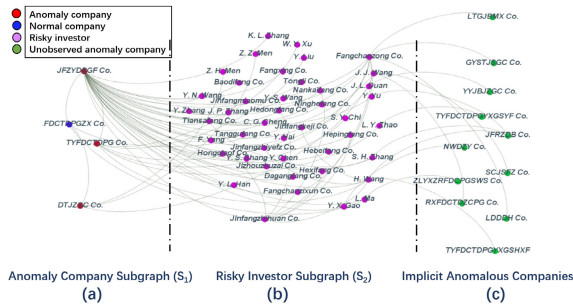


Fig. 8. Risky investor detection in Nov 2017. (a) Connected anomaly company subgraph S_1 includes 3 anomalous microcredit loan companies with observed anomalous business activities (red dots) and 1 normal company (a blue dot). (b) Disconnected alignment anomaly subgraph S_2 consists of 44 detected risky investors (violet dots). (c) 11 companies with unobserved abnormal business behaviors (green dots) related to risky investors, discovered from the ground truth.

and 1814 relationships, and G_2 with 4785 shareholders and 35 892 coinvestment links. Twenty four months of business data (January 2017–December 2018) are used to compute anomaly features for each company. In November 2017, four interconnected companies (three anomalous, one normal) are detected [Fig. 8(a)], and 44 risky investors are identified [Fig. 8(b)]. To validate our approach’s accuracy, we identified 11 additional companies where 43 of these investors had made investments, as shown in Fig. 8(c). This proves that these investors have a certain degree of business risk.

3) *Anomaly Event Detection-Epidemic Breakout*: In this scenario, our objective is to illustrate the strong association between migration patterns and the COVID-19 outbreak. To comprehend the causal chain behind the COVID-19 epidemic breakout, we analyze several Chinese cities characterized by significant population movements and a high number of COVID-19 cases. We utilize the Baidu migration graph as our source graph G_1 , and consider COVID-19 as the target graph G_2 . This setup is designed to simulate the inherently unpredictable nature of COVID-19 transmission. During the initial 23 days of January 2020, we identified significant anomalous migration subgraphs over a span of 14 days. These subgraphs revealed a similar regional distribution pattern among the cities involved, as illustrated in Fig. 9(a). For instance, when Wuhan initially reported confirmed COVID-19 cases on January 13, 2020, a substantial increase in population movements occurred between 367 cities in Mainland China. After 6 days, our

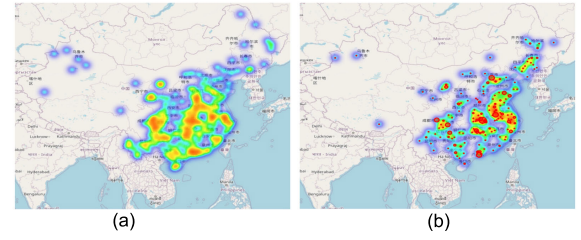


Fig. 9. Intuitive interpretation of the cause of the anomaly event: epidemic breakout. (a) Cities with significant population movements on January 13, 2020. (b) Anomalous COVID-19 outbreak cities detected by IASD from January 19, 2020 to February 20, 2020.

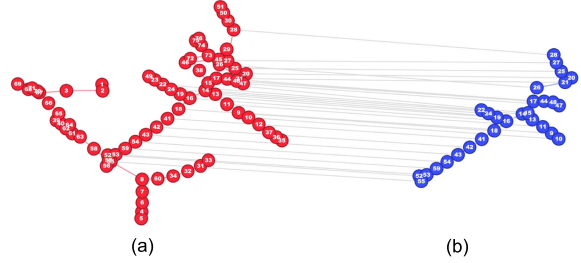


Fig. 10. Water pollution prediction Case Study. Blue circles indicate sensors that will issue water pollution warnings in the upcoming time slice, and red circles show the current warning sensors detected at the 3th time slice.

approach identifies the first anomaly subgraph S_2 within the epidemic graph G_2 , leveraging the transferred migration attributes. This discovery underscores our approach’s capability in epidemic detection. Subsequently, IASD proceeds to detect an additional 32 anomaly subgraphs over the following 33 days, covering a total of 157 cities that were officially confirmed to have experienced new case outbreaks according to Chinese government news websites. These cities are marked on the map according to their geographic coordinates, as shown in Fig. 9(b). This experiment offers persuasive evidence of a congruent distribution between migration patterns and the COVID-19 outbreak. It offers an intuitive explanation for the rapid spread of the COVID-19 epidemic, largely attributed to population migration.

4) *WP Prediction*: IASD can be applied to a set of graphs with time series, e.g., WSN across different time slices. We use two time-slice graphs $G^{(t)}$ with historical sensors’ data and $G^{(t+1)}$ without any sensors’ warning data, to construct the IASD’s two-layer graph. i.e., $G^{(t)}$ is G_1 , and $G^{(t+1)}$ is G_2 . In Fig. 10, IASD identified 76 sensors emitting warning signals during the third time slot and determined that 29 sensors would predict warnings of chemical contaminants in the fourth time slot. These 29 sensors had not previously emitted a warning signal; however, 27 of them generated actual warnings in the fourth time slice, and in the fifth time slice, two additional sensors also issued warning signals. This experiment demonstrates that IASD is capable of effectively addressing the practical task of predicting WP.

VI. CONCLUSION

In this article, we propose the *anomaly alignment* problem and a novel anomaly detection algorithm IASD for the multilayer multidimensional attributed graph. IASD employs network alignment and a multilayer graph model to integrate a feature-rich source graph with the target graph, facilitating the detection of IASs. Various evaluation across five real datasets demonstrates IASD’s effective performance and robustness.

In future work, under the Anomaly Alignment framework, we aim to extend IASD to more complex multilayer graphs by aligning multiple source graphs and transferring features across layers, using iterative or joint alignment strategies, and validating the method on diverse multilayer datasets, such as SNs and multiomics networks.

APPENDIX A DATASETS

We provide detailed information about the six real datasets used in our experiments. The raw data used to construct our multilayer graphs are processed in the following steps.

A. Weibo Dataset

The Weibo dataset includes 5.76 million tweets and 1.81 billion retweets collected from the Weibo Website from December 1, 2011 to December 31, 2012. All these tweets are assigned a geo-tag attribute which is a longitude and latitude within the urban area of Tianjin (*The latitudes are in the range of 37°89' to 40°77', and the longitudes are in the range of 114°13' to 120°64'*). We extracted 17.26 million users and their relationships from these tweets, analyzed the semantics of these tweets based on the SentiCircle algorithm [44], and classified the tweets into three categories: negative, positive, and neutral sentiment tweets. 99,872 negative sentiment tweets containing negative emotional words (e.g., “hate,” “complain,” etc.) are filtered to construct the source graph G_1 (*Negative Friendship Graph*). G_1 includes 11 225 Weibo users and 13 405 friend (or follower) relationships.

The target graph G_2 is the *Neighborhood Graph* which includes 462 591 geo-coordinates extracted from the tweets. These geo-coordinates are clustered into 137 919 vertices. Each vertex represents a *neighborhood* (a circular area with a diameter of 300 m), and the 6 396 462 edges are extracted from all those negative tweets. Each edge represents that there exists a user who posted tweets with the geo-coordinates of two neighborhoods.

Ground Truth Data: The authority department’s urban security database provides 3.21 million security records of urban and rural areas, which constitute the ground truth for Urban Crime Hotspot Detection. All security alarm events occurring within the same time and region as the Weibo data are extracted. The data items include the time, location, and alarm information. Its location is labeled by longitude and latitude coordinates. Thus, our detection result is deemed correct if there is an alarm record within the hotspot area (i.e., a node of S_2) identified by the algorithm; otherwise, it is deemed incorrect.

B. TYC Dataset

This dataset is extracted from the website www.tianyancha.com. It includes 1 668 797 corporations, 1 000 938 investors, and 2 657 470 of their investment-relationships during the period from January 1, 2000 to December 31, 2018. We collect 332 009 corporations and their obvious abnormal commercial-related behaviors. For example, their assets are auctioned off by the court, or they announce being unable to repay their bank loans, or they (or their legal persons) are accused, and so on. In this case study, we focus on 3005 microcredit loan companies and 8146 related shareholders (including persons and companies) from January 1, 2017 to December 31, 2018.

Therefore, G_1 has 864 microcredit loan companies as vertices and 1814 associated relationships between these companies as edges. G_2 has 4785 related shareholders as vertices and 35 892 co-investment relationships between these shareholders as edges.

Ground Truth Data: As the website designates companies as high-risk by compiling records of warnings, penalties, and actions issued by courts or regulatory authorities throughout their operations, we employed this labeled information as the ground truth, which encompasses both the identified companies and their corresponding shareholders.

C. Baidu Migration (B) Dataset Versus Covid-19 Pandemic (C) Dataset

The Baidu migration dataset includes 4.8 million population movement records among 441 Chinese cities collected from the *Baidu.com* website during the period from January 1, 2020 to February 29, 2020. The population movement records are the immigration and emigration rates of each city. Therefore, G_1 is constructed with 441 cities and the edges which represent the top 100 population movement relationships between a city and its neighboring cities.

The COVID-19 pandemic dataset is collected from GitHub (December 1, 2019–January 27, 2020) and the Chinese News Websites (January 27, 2020–March 10, 2020). It includes 24,787 confirmed cases of COVID-19 reported daily and 7,686 known patient trajectories announced as of March 10, 2020. Those cases were distributed in 313 cities in Mainland China. We construct G_2 as a plain graph². G_2 consists of 313 cities as vertices and 7686 known patient trajectories as edges. Here, our approach only adopts the topology of G_2 without using any Covid-19 outbreak information as the attributes.

Ground Truth Data: All COVID-19 confirmed case data collected from the Chinese News Websites are merely used as ground truth to evaluate our algorithm.

D. WP Dataset

This WP warning data was collected from a real water sensor network³ across six time slices. The dataset comprises 12 527 sensors (nodes) and 14 831 edges. These sensors are responsible for monitoring chemical pollutants distributed in river areas. A sensor produces the signal ‘1’ when there is the presence of water contamination; otherwise, it reports ‘0’. In our experiment, the graphs G_1 and G_2 are the largest connected subgraphs within this dataset. The source graph G_1 is built based on the structure and attributes of the network at the t time slice, while the target graph G_2 is constructed from the structure of the network at the $t + 1$ time slice.

Ground Truth Data: In our article, the sensor alarm signals from the corresponding time slices in the WP dataset are used as the ground truth for our detection time slices.

E. Synthetic Dataset

Synthetic dataset is constructed with reference to the basic characteristics of the Weibo dataset, such as the proportion of nodes and the proportion of aligned edges, as detailed in Table V. For constructing individual graphs, we adopted the Barabasi-Albert model [45], and the feature data were simulated based on the basic information from the Weibo dataset.

²The plain graph refers to the graph without any attribute feature.

³“The Battle of the Water Sensor Networks (BWSN)”.

TABLE IV
STATISTICS OF OUR MULTILAYER GRAPHS

Multilayer graph	Graph	#Nodes	#Edges	Degree	Diameter	Density	Modularity	Path length	Clustering coefficient	#Communities
Weibo	G_1	300	310	1.03	5	0.003	0.88	1.85	0.015	15
	G_2	1184	10971	9.27	12	0.008	0.85	3.89	0.457	23
Water Pollution	G_1	1,880	2,102	2.24	107	0.001	0.99	42.29	0.01	2
	G_2									
Baidu & Covid-19	G_1	298	43,353	145.48	2	0.49	0.005	1.021	0.49	2
	G_2	313	2,185	6.98	8	0.022	0.31	2.59	0.29	7
TYC	G_1	864	1,814	2.10	12	0.002	0.83	3.58	0.21	28
	G_2	4,785	35,892	7.50	15	0.002	0.90	5.02	0.44	47

APPENDIX B THEORY

A. Proposition

Proposition 2 (Anomaly Alignment): Given a two-layer graph $N = (G_1, G_2, M)$, where $G_1 = (V_1, E_1, X_{G_1})$ is the source graph and $G_2 = (V_2, E_2, X_{G_2})$ is the target graph, with alignment edges $M \subseteq V_1 \times V_2$. A connected anomaly subgraph $S_1(V_{S_1}, E_{S_1}, X_{S_1})$ in G_1 contains an anomalous node v_i . If $(v_i, v_j) \in E_{S_1}$ and $(v'_i, v'_j) \in E_2$ exist with $(v_i, v'_i) \in M$ and $(v_j, v'_j) \in M$, then according to our feature transfer learning model $f_{\text{tran}} : X_{S_1} \rightarrow X_{S_2}$, a nonempty aligned anomaly subgraph S_2 exists in G_2 .

Proof: Let $\mathbf{s}_1^{n \times 1}$ be the indicator vector of subgraph S_1 , $\mathbf{s}_2^{m \times 1}$ the indicator vector of S_2 , and $\mathbf{M}^{m \times n}$ the alignment matrix. The attribute matrices of subgraph S_1 and S_2 are defined as

$$X_{S_1} = \text{diag}(\mathbf{s}_1) X_{G_1}, \quad X_{S_2} = \text{diag}(\mathbf{s}_2) X_{G_2} \quad (19)$$

where $\text{diag}(\mathbf{s}_1) \in \mathbb{R}^{n \times n}$ denotes the diagonal matrix with the diagonal entries from $\mathbf{s}_1$. Similarly, $\text{diag}(\mathbf{s}_2) \in \mathbb{R}^{m \times m}$ is the diagonal matrix with the diagonal entries from $\mathbf{s}_2$.

If $(v_i, v_j) \in E_{S_1}$, the attribute matrices of subgraphs S_1 and S_2 are

$$X_{S_1} = \begin{bmatrix} x_{S_1}(v_i) \\ x_{S_1}(v_j) \end{bmatrix}, \quad X_{S_2} = \begin{bmatrix} x_{S_2}(v'_i) \\ x_{S_2}(v'_j) \end{bmatrix} \quad (20)$$

where X_{S_1} includes the anomalous attribute $x_{S_1}(v_i)$, while X_{S_2} contains only normal attributes.

Since v_i aligns with v'_i and v_j aligns with v'_j , we have $\mathbf{M}(i', i) = 1$ and $\mathbf{M}(j', j) = 1$. Based on the definition of our feature transfer learning model: $f_{\text{tran}} : X_{S_1} \rightarrow X_{S_2}$, the projection of the transferred features onto X_{S_2} are represented using the following equation:

$$X_{S_2} = \begin{bmatrix} x_{S_2}(v'_i) + \Phi(x_{S_1}(v_i)) \\ x_{S_2}(v'_j) + \Phi(x_{S_1}(v_j)) \end{bmatrix}. \quad (21)$$

Thus, X_{S_2} includes the transferred anomaly features, and the aligned subgraph $S_2 = \{v'_i, v'_j\}$ is an anomaly subgraph. The proposition is thereby proven. $\square$

B. Lemma

Lemma 5 [7]: Given $\bar{N}_\alpha(S_1) = N(S_1) - N_\alpha(S_1)$, the optimal subgraph of NPGS problem with the K -budget node-weighted prize-collection Steiner Tree is defined as

$$S_1^{\alpha, K} = \arg \max_{S_1 \subseteq G_1, S_1 \text{ is connected}} N_\alpha(S_1), \quad \text{s.t. } \bar{N}_\alpha(S_1) \leq K \quad (22)$$

where α is the anomaly significance level, typically set as $\alpha \leq 0.15$ in this article. $K \leq n$ is the number of normal nodes in S_1 .

Lemma 6 [7]: Given $S_1^{\alpha, K}$, ($K = 0, \dots, n$) is the optimal solution of the K -budget node-weighted prize-collection Steiner Tree from (22), the NPGS problem

$$\hat{S}_1 = \arg \max_{S_1 \subseteq G_1, S_1 \text{ is connected}} F_\alpha(S_1)$$

is equivalent to the following optimal problem:

$$\hat{S}_1 = \arg \max_{\alpha \in \mathbb{P}, S_1 \in \{S_1^{\alpha, 0}, \dots, S_1^{\alpha, n}\}} F_\alpha(S_1). \quad (23)$$

Lemma 7: Optimal anomaly S_2 with S_1 and $\mathbf{M}$ fixed. Under the condition that the anomaly subgraph S_1 and the alignment matrix $\mathbf{M}$ are fixed, if S_1 and S_2 satisfy anomaly alignment (Proposition 2), then the maximum solution $\hat{S}_2$ of function Q is defined as follows:

$$\hat{S}_2 = \lambda \mathbf{M} [\mathbf{s}_1 \odot (\mathbf{1} - \mathbf{x}_1)] + (\lambda \mathbf{x}_2^T + 1) \mathbf{M} \mathbf{s}_1 \quad (24)$$

where λ is a regularization parameter.

Proof: Because the anomaly alignment property holds, $\mathbf{s}_1 \neq 0$ and $M \neq 0$. If $\mathbf{s}_1$ and $\mathbf{M}$ are fixed, then our feature transfer learning function is defined as follows:

$$FT(S_1, S_2, M) = \frac{\mathbf{s}_2^T M [\mathbf{s}_1 \odot (\mathbf{1} - \mathbf{x}_1)] + (\mathbf{s}_2^T \odot \mathbf{x}_2^T) \mathbf{M} \mathbf{s}_1}{\|\mathbf{1} - \mathbf{x}_1\|_1 \cdot \|\mathbf{x}_2\|_1}. \quad (25)$$

To enhance the stability of the model and improve its generalization ability, (26) reformulates the above equation by introducing an additional regularization term

$$FT(\mathbf{s}_2) = \frac{\mathbf{s}_2^T \mathbf{M} [\mathbf{s}_1 \odot (\mathbf{1} - \mathbf{x}_1)] + (\mathbf{s}_2^T \odot \mathbf{x}_2^T) \mathbf{M} \mathbf{s}_1}{\|(1 - \mathbf{x}_1)\|_1 \cdot \|\mathbf{x}_2\|_1} - \rho \|\mathbf{s}_2 - \mathbf{M} \mathbf{s}_1\|_F^2 \quad (26)$$

where ρ is a regularization parameter. Let the derivative be equal to zero. Since the denominator of (26) is a constant value, then

$$\frac{\partial FT(\mathbf{s}_2)}{\partial \mathbf{s}_2} = \frac{\partial}{\partial \mathbf{s}_2} (\mathbf{s}_2^T \mathbf{M} [\mathbf{s}_1 \odot (\mathbf{1} - \mathbf{x}_1)] + (\mathbf{s}_2^T \odot \mathbf{x}_2^T) \mathbf{M} \mathbf{s}_1 - \rho \|\mathbf{s}_2 - \mathbf{M} \mathbf{s}_1\|_F^2) = 0. \quad (27)$$

The maximum solution of the induced subgraph feature scoring function is obtained as follows:

$$\hat{\mathbf{s}}_2 = \frac{1}{2\rho} \{\mathbf{M} [\mathbf{s}_1 \odot (\mathbf{1} - \mathbf{x}_1)] + \mathbf{x}_2^T \mathbf{M} \mathbf{s}_1\} + \mathbf{M} \mathbf{s}_1. \quad (28)$$

Let $\lambda = 1/2\rho$, then

$$\hat{\mathbf{s}}_2 = \lambda \mathbf{M} [\mathbf{s}_1 \odot (\mathbf{1} - \mathbf{x}_1)] + (\lambda \mathbf{x}_2^T + 1) \mathbf{M} \mathbf{s}_1. \quad (29)$$

Therefore, the lemma is proved. $\square$

Lemma 8: Optimal alignment matrix $\mathbf{M}$ with S_1 and S_2 fixed. Given two input subgraphs S_1 and S_2 , then Function Q in (16) w.r.t. $\mathbf{M}$ is equivalent to the optimal problem.

$$\hat{\mathbf{M}} = \arg \max_{\mathbf{M} \in [0,1]^{m \times n}} \eta [Q(\mathbf{M})] - (1 - \eta) \|\mathbf{M} - \mathbf{H}\|_F^2 \quad (30)$$

where η is a regularization parameter, $\mathbf{H}$ is the prior alignment matrix, and $\|\cdot\|_F$ denotes the Frobenius norm.

Proof: Given the input graphs S_1 and S_2 , function $Q(S_1, S_2, \mathbf{M})$ is equivalent to the function w.r.t. $\mathbf{M}$ in (31). To simplify the derivation of formulas, we introduce the vector $\mathbf{m} = \text{vec}(\mathbf{M}) \in \mathbb{R}^{mn \times 1}$, where $\text{vec}(\cdot)$ denotes the vectorization operator that stacks the columns of a matrix sequentially

$$Q(\mathbf{M}) = \eta [ICS(\mathbf{M}) + FT(\mathbf{M})] + (1 - \eta) \|\mathbf{M} - \mathbf{H}\|_F^2. \quad (31)$$

The maximum solution of it is deduced by setting its derivative to be zero

$$\begin{aligned} \frac{\partial Q(\mathbf{m})}{\partial \mathbf{m}} &= \eta \left[A_{S_1} \otimes A_{S_2} + (A_{S_1} \otimes A_{S_2})^T \right] \mathbf{m} - 2(1 - \eta) \mathbf{m} \\ &\quad + \eta \left[\mathbf{s}_2 (\mathbf{s}_1^T \odot (\mathbf{1} - \mathbf{x}_1)^T) + \mathbf{x}_2 \mathbf{s}_1^T \right] - 2(1 - \eta) \mathbf{h} = 0 \end{aligned}$$

then the alignment matrix $\mathbf{M}$ is calculated as follows:

$$\begin{aligned} \hat{\mathbf{m}} &= \eta \left[A_{S_1} \otimes A_{S_2} + (A_{S_1} \otimes A_{S_2})^T \right] \mathbf{m} \\ &\quad + \eta \left[\mathbf{s}_2 (\mathbf{s}_1^T \odot (\mathbf{1} - \mathbf{x}_1)^T) + \mathbf{x}_2 \mathbf{s}_1^T \right] - \mathbf{h} \\ &= \eta \left[(A_{S_1} \otimes A_{S_2}) \mathbf{m} + (A_{S_1}^T \otimes A_{S_2}^T) \mathbf{m} \right. \\ &\quad \left. + \eta \left[\mathbf{s}_2 (\mathbf{s}_1^T \odot (\mathbf{1} - \mathbf{x}_1)^T) + \mathbf{x}_2 \mathbf{s}_1^T \right] - \mathbf{h} \right]. \end{aligned}$$

Then, we apply the nature of the Kronecker product: $(B^T \otimes A) \text{vec}(X) = \text{vec}(AXB)$ [46] to make our algorithm more efficient

$$\begin{aligned} \hat{\mathbf{m}} &= \eta \left[\text{vec}(A_{S_2} \mathbf{M} A_{S_1}^T) + \text{vec}(A_{S_2}^T \mathbf{M} A_{S_1}) \right] \\ &\quad + \eta \left[\mathbf{s}_2 (\mathbf{s}_1^T \odot (\mathbf{1} - \mathbf{x}_1)^T) + \mathbf{x}_2 \mathbf{s}_1^T \right] - \mathbf{h}. \quad (32) \end{aligned}$$

Consequently, the maximum solution of the function $Q(\mathbf{M})$ exists, and $\hat{\mathbf{m}}$ in (32) is the fixed-point iterative method. The lemma is proved. $\square$

In a given real-world scenario, G_1 and G_2 represent a pair of associated graphs. If an associated link (v_i, v_j) exists between G_1 and G_2 , where $v_i \in V_1$ and $v_j \in V_2$, then $H(i, j) = 1$; otherwise, $H(i, j) = 0$. In the iterative optimal procedure, when S_1 and S_2 are provided, $Q(\cdot)$ is solely dependent on $\mathbf{M}$. The regularization term $\|\mathbf{M} - \mathbf{H}\|_F^2$ in (30) would help prevent a trivial solution of (30), while $\mathbf{M}$ is equal to the zero matrix.

Additionally, the time and space complexities of computing the Kronecker products $A_{S_1} \otimes A_{S_2}$ and $A_{S_1}^T \otimes A_{S_2}^T$ are both $O(n^2 m^2)$, whereas the vectorized forms $\text{vec}(A_{S_2} \mathbf{M} A_{S_1}^T)$ and $\text{vec}(A_{S_2}^T \mathbf{M} A_{S_1})$ only required $O(mn^2)$. By exploiting this property of the Kronecker product, we optimize this Iterative process and reduce its overall complexity to $O(mn^2)$.

TABLE V
RUNNING TIME ANALYSIS ON SYNTHETIC DATASET

G_1		G_2		Associated links	Time ($\times 10^3$ s)
Nodes	Edges	Nodes	Edges		
100	300	196	596	1,600	3.93
200	600	396	1,196	3,200	8.96
300	900	596	1,796	4,800	18.52
400	1,200	796	2,396	6,400	37.40
500	1,500	996	2,996	8,000	64.02
600	1,800	1,196	3,596	9,600	99.46
700	2,100	1,396	4,196	11,200	135.41
800	2,400	1,596	4,796	12,800	172.45
900	2,700	1,796	5,396	14,400	252.46
1,000	3,000	1,996	5,996	16,000	350.90

C. Theorem

Theorem 2: Optimal anomaly subgraphs S_1, S_2 in Graphs G_1, G_2 . If subgraphs S_1 and S_2 satisfy the anomaly alignment condition, then in the IASD problem [5], an optimal solution exists and is defined as follows:

$$(\hat{S}_1, \hat{S}_2) = \arg \max_{\alpha \in \mathbb{P}} F_\alpha(S_1) + Q(S_1, S_2, M). \quad (33)$$

Proof: Based on the definition of our IASD problem and Lemma 5, S_1 contains at most K normal nodes, while S_2 consists of at most c nodes. Both S_1 and S_2 are therefore restricted subgraphs. If α is assigned a specific value, then the functions $F_\alpha(S_1)$ and $Q(S_1, S_2, M)$ are guaranteed to be monotonically increasing, as ensured by properties P1, P2, and P3. Consequently, if S_1 and S_2 satisfy the Anomaly alignment condition, then the optimal solution to the IASD problem, (2), exists.

The uniqueness of Theorem 1 is proved by means of contradiction. Suppose $(\hat{S}_1, \hat{S}_2)$ is not the unique optimal solution of problem [2]. We assume that there exists at least one pair of nodes (v_i, v'_i) aligned, where $v_i \notin \hat{S}_1$ but is connected to $\hat{S}_1$, $v'_i \notin \hat{S}_2$, the P-value of the node v_i 's feature is less than a significant level $p(\mathbf{x}_1(\mathbf{t})) \leq \alpha$, and the edge (v'_i, v_i) exists, i.e., $\mathbf{M}(t', t) > 0$. Let $S_1' = \hat{S}_1 \cup \{v_i\}$ and $S_2' = \hat{S}_2 \cup \{v'_i\}$. Since $N_\alpha(\cdot)$ denotes the number of anomaly nodes, then $N_\alpha(S_1') > N_\alpha(\hat{S}_1)$ holds. By properties (P1), (P2), it follows that $F(S_1') \geq F(\hat{S}_1)$. Moreover, by properties (P1), (P3), $Q(S_1', S_2', M)$ increases monotonically with both $\sum_{v_i \in V_{S_1}', v'_i \in V_{S_2}'} \mathbf{M}(i', i)$ and $N_\alpha(S_1')$. Let $Q' = Q(S_1', S_2', M)$ and $\hat{Q} = Q(\hat{S}_1, \hat{S}_2, M)$. Then $Q' > \hat{Q}$, which implies $F_\alpha(S_1') + Q' > F_\alpha(\hat{S}_1) + \hat{Q}$.

This assumption contradicts that $F_\alpha(\hat{S}_1) + \hat{Q}$ is the maximum value of the IASD problem [2]. Therefore, the pair of nodes (v_i, v'_i) cannot exist, and it follows that $(\hat{S}_1, \hat{S}_2)$ is the optimal solution to the IASD problem. Hence, the theorem is proven. $\square$

APPENDIX C EXPERIMENTAL RESULTS

A. Evaluation on the HC Statistic

As shown in Table VI, the results of IASD(HC) and IASD(BJ) are highly similar. In conjunction with the comparison against TF(HC) and A3MAN(HC), these findings further substantiate the effectiveness of the proposed IASD approach.

B. Parameter Sensitivity Study

- 1) Influence of the Anomaly Ratio of G_1 : The anomaly ratio refers to the proportion of anomalous vertices in

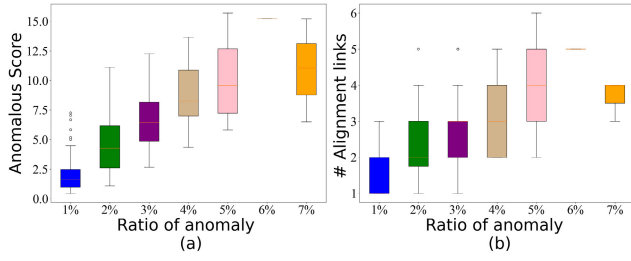


Fig. 11. Influence of the anomalous ratio of G_1 . When the ratio of anomalous nodes in source graph G_1 increases from 0.5% ~ 7%. (a) Anomalous scores of S_2 (i.e., the induced anomaly features) increase with the growing anomaly ratio in S_1 . (b) Number of alignment links between S_1 and S_2 follows a rising trend, reaching a peak before declining.

G_1 . We evaluate the influence of the anomaly ratio by detecting the anomaly features of every vertex in G_1 . As the anomaly ratio increases from 0.5% to 5%, the anomaly feature score of S_2 , induced by transfer learning, increases from 0 to 15. Additionally, the number of alignment links calculated by our approach increases monotonically with the anomaly ratio in G_1 , as shown in Fig. 11. However, when the anomaly ratio of G_1 exceeds 5%, the anomalous scores of S_2 remain stable, while the number of alignment links begins to decrease. Hence, this experiment demonstrates that, within an anomaly alignment structure, the anomaly features of the source graph can effectively amplify those in the target graph, thereby facilitating the detection of implicit anomalous subgraphs. Furthermore, this experiment validates the reasonableness of the properties (P1, P2, and P3) proposed in our problem definition and demonstrates the effectiveness of our approach.

- 2) Influence of the Parameter c : As defined in (2), $|V_{S_2}| \leq c$, the number of S_2 's nodes is constrained by a constant c , where $c \leq |V_2|$. For ease of visualization, we represent the parameter c as $|V_{S_2}|/|V_2|$, that is the ratio of the number of nodes in S_2 to the total number of nodes in G_2 . In Fig. 12, we investigate c within the range of (0, 11.2%] in the WP dataset to assess our algorithm's performance. For $c \leq 3.2\%$, Precision, Recall, $F1$, Accuracy, TPR-P, and TPR-D are all equal to 1, and FPR, FPR-P, and FPR-D are all 0. As c falls within [3.2%, 11.2%], Precision remains 1, and FPR, FPR-P, and FPR-D are 0; however, Recall, $F1$, Accuracy, and TPR-P (similar to TPR-D) exhibit a declining trend, as illustrated in Fig. 12.
- 3) Influence of the Parameter K : According to [7] and [8], function $F_\alpha(S_1)$ increases with the number of anomaly nodes of S_1 , and tends to stabilize once the number of normal nodes K in S_1 exceeds 20. That is, $F_\alpha(S_1)$ achieves the highest statistic score when $K \geq 20$. Since our algorithm employs TSPSD [7] to detect anomaly subgraph S_1 , the parameter K in TSPSD similarly affects the detection performance of our IASD. As shown in Fig. 13, IASD generally exhibits superior detection performance when $K = 20$ in our experiments. Therefore, we typically tune the parameter K to be equal 20.
- 4) Influence of the Parameter $\alpha_{\max}$: In the method proposed in this article, the setting of the parameter $\alpha_{\max}$ is of critical importance to the model's performance. As depicted in Fig. 14, we conducted a systematic analysis of the model's performance on the TYC dataset with

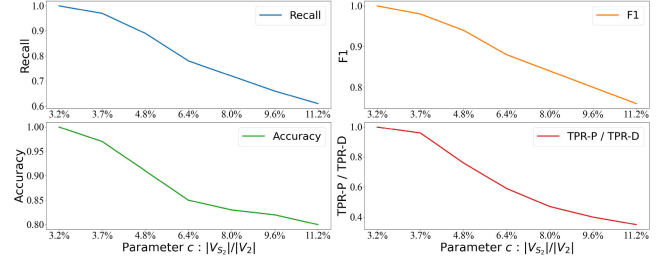


Fig. 12. Influence of the parameter c : (Ratio of the number of nodes in S_2 to the total nodes in WP Graph G_2).

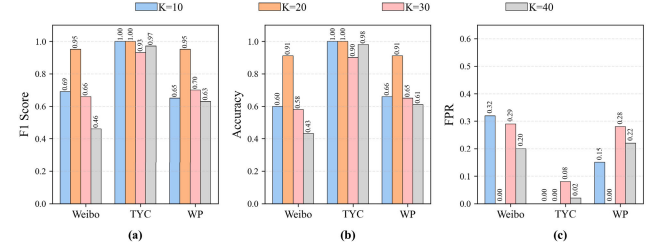


Fig. 13. Effect of parameter K on the detection performance of IASD. (a) $F1$ Score. (b) Accuracy. (c) FPR.

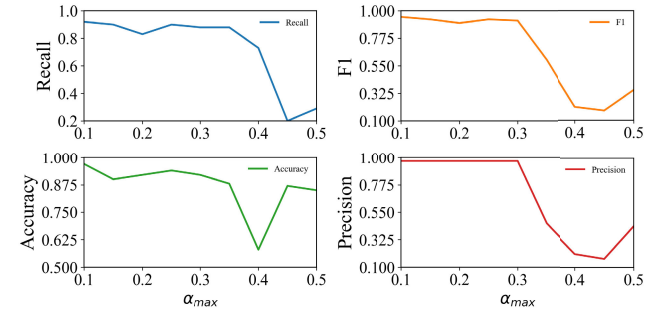


Fig. 14. Influence of the parameter $\alpha_{\max}$ on the TYC dataset.

respect to four key metrics: Recall, $F1$ -score, Accuracy, and Precision, across various values of $\alpha_{\max}$. The experimental results indicate that optimal model performance is achieved when the value of $\alpha_{\max}$ falls within a specific range. Specifically, when $\alpha_{\max}$ is within the range [0.10, 0.35], the model's Recall, Accuracy, and Precision metrics are maintained at a high level. Notably, both Accuracy and Precision metrics peak around $\alpha_{\max} = 0.25$. This suggests that within this parameter interval, the model can effectively balance the comprehensiveness and accuracy of its identification. However, when the value of $\alpha_{\max}$ exceeds 0.35, the model's performance metrics begin to decline sharply. Particularly for $\alpha_{\max} \geq 0.45$, both Recall and the $F1$ -score decrease significantly, indicating that an excessively large $\alpha_{\max}$ value substantially weakens the model's ability to identify positive samples, thereby severely compromising its overall performance.

- 5) Influence of the GAT Head h : As depicted in Fig. 15, the number of GAT heads h significantly influences performance. We investigate the impact of the number of attention heads on the performance of a Transformer-based model. Our analysis reveals a nonmonotonic relationship between the number of heads (h) and model performance. Specifically, a single attention head ($h = 1$) yields suboptimal performance, sug-

TABLE VI
IASD (HC), TF (HC), AND A3MAN (HC) ON WP, WEIBO, AND TYC DATASET

Dataset	Method	Accuracy	Recall	F1	Precision	FPR	TPR-P	TPR-D	FPR-P	FPR-D
Weibo	IASD(HC)	0.91	0.91	0.95	1.00	0.00	0.91	0.89	0.00	0.00
	TF(HC)	0.88	0.88	0.90	0.94	0.10	0.88	0.88	0.22	0.24
	A3MAN(HC)	0.36	0.28	0.37	0.70	0.30	0.29	0.28	0.19	0.28
TYC	IASD(HC)	0.90	0.90	0.93	0.97	0.08	0.90	0.90	0.08	0.08
	TF(HC)	0.60	0.60	0.75	1.00	0.00	0.60	0.60	0.00	0.00
	A3MAN(HC)	0.50	0.44	0.19	0.15	0.49	0.35	0.51	0.74	0.21
WP	IASD(HC)	0.83	0.82	0.90	1.00	0.00	0.78	0.78	0.00	0.00
	TF(HC)	0.75	0.75	0.82	1.00	0.00	0.61	0.61	0.49	0.49
	A3MAN(HC)	0.16	0.07	0.13	0.87	0.09	0.10	0.10	0.01	0.01

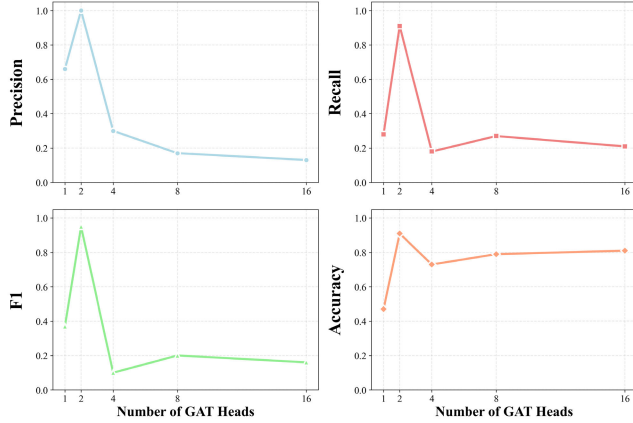


Fig. 15. Influence of GAT Head h on the WP dataset.

gesting limitations in capturing the diverse feature representations necessary for robust modeling. Increasing the number of heads to $h = 2$ results in a substantial performance improvement, which validates the efficacy of the multihead attention mechanism in learning distinct, complementary representations. However, a critical observation is the sharp performance degradation when the number of heads becomes excessively large. Our findings underscore the importance of judiciously selecting the number of attention heads to balance representational power with the risk of overfitting.

C. Running Time Analysis

To evaluate the time complexity, a synthetic dataset was constructed with reference to the basic characteristics of the Weibo dataset, including the proportion of nodes and aligned edges, as summarized in Table V. For the generation of individual graphs, we employed the Barabási–Albert model [45], while the feature data were simulated using the basic information from the Weibo dataset. Ten sets of simulation graphs were generated for empirical evaluation, as reported in Table V, which demonstrate that the algorithm’s running time grows proportionally with the cube of the number of graph nodes, i.e., $O(n^3)$.

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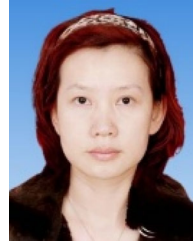
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Nannan Wu received the B.S. degree from Zhengzhou University, Zhengzhou, China, in 2009, the M.S. degree from Chongqing University, Chongqing, China, in 2012, and the Ph.D. degree in computer software and theory from Beihang University, Beijing, China, in 2018.

He is a Professor with Tianjin University, Tianjin, China. His current research interests include artificial intelligence and graph anomaly detection in large-scale graphs.



Ying Sun received the B.S. degree in computer science and engineering from Tianjin University, Tianjin, China, in 1990, the M.S. degree in control science and engineering from Xi'an Jiaotong University, Xi'an, China, in 1998, and the Ph.D. degree from Tianjin University in 2025.

She is currently a Professor with the School of International Business, Zhejiang Yuexiu University, Shaoxing, China. Her research interests include anomaly detection, human behavior dynamics analysis by complex networks, and large-scale data mining.



Yazheng Zhao received the B.S. degree from Shanghai University of Finance and Economics, Shanghai, China, in 2024. He is currently pursuing the M.S. degree with the School of Computer Science and Technology, Tianjin University, Tianjin, China.

His current research interests include graph mining and graph anomaly detection.



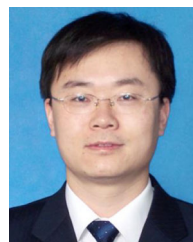
Wenjun Wang received the Ph.D. degree from Peking University, Beijing, China, in 2006.

He is currently a Professor with the College of Intelligence and Computing, Tianjin University, Tianjin, China, the Chief Expert of major projects of the National Social Science Foundation, Beijing, the Big Data Specially Invited Expert with Tianjin Public Security Bureau, Tianjin, and the Director of Tianjin Engineering Research Center of Big Data on Public Security, Tianjin. His research interests include computational social science, large-scale data mining, intelligence analysis, and multilayer complex network modeling.



Xueli Liu received the Ph.D. degree from Harbin Institute of Technology, Harbin, China, in 2018.

She is an Associate Professor with the School of Computer Science and Technology, Tianjin University, Tianjin, China. Her research interests include big data computing, graph association analysis, and deep learning interpretation.



Jianxin Li is currently a Professor with the State Key Laboratory of Software Development Environment and Beijing Advanced Innovation Center for Big Data and Brain Computing, Beihang University, Beijing, China. He has published 30 refereed articles in major data mining venues, including IEEE TRANSACTIONS ON KNOWLEDGE AND DATA ENGINEERING, ACM-KDD, ACM-WWW, AAAI and IJCAI. His current research interests include machine learning, distributed system, trust management and network security.

Mr. Li won the best paper award at AAAI2021.